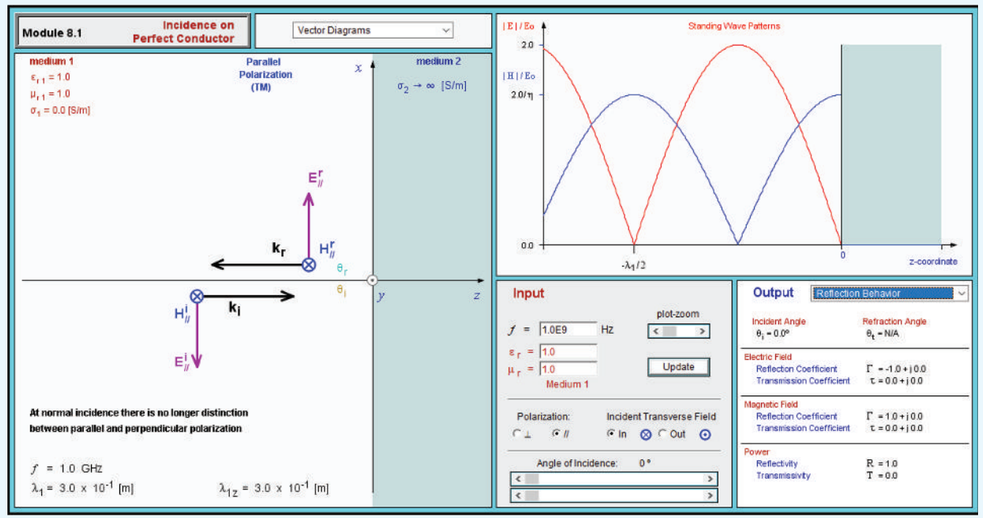
Module 8.1 Incidence on Perfect Conductor Observe the standing-wave pattern created by the combination of a wave incident normally onto the plane surface of a conductor and its reflection.



Concept Question 8-3: Explain on the basis of boundary conditions why it is necessary that $\Gamma = -1$ at the boundary between a dielectric and a perfect conductor.

Answer: For incidence in medium 1 $(\varepsilon_1, \mu_0, \sigma_1)$ onto medium 2 $(\varepsilon_2, \mu_0, \sigma_2)$,

$$\Gamma = \frac{\sqrt{\varepsilon_{c_1}} - \sqrt{\varepsilon_{c_2}}}{\sqrt{\varepsilon_{c_1}} + \sqrt{\varepsilon_{c_2}}},$$

with $\varepsilon_{c_1} = (\varepsilon_1 - j\sigma_1/\omega)$ and $\varepsilon_{c_2} = (\varepsilon_2 - j\sigma_2/\omega)$. (See EM.)

Exercise 8-1: To eliminate reflections of normally incident plane waves, a dielectric slab of thickness d and relative permittivity ε_{r_2} is to be inserted between two semi-infinite media with relative permittivities $\varepsilon_{r_1} = 1$ and $\varepsilon_{r_3} = 16$. Use the quarter-wave transformer technique to select d and ε_{r_2}. Assume $f = 3$ GHz.

Answer: $\varepsilon_{r_2} = 4$ and $d = (1.25 + 2.5n)$ (cm), with $n = 0, 1, 2, \ldots$. (See EM.)

Exercise 8-2: Express the normal-incidence reflection coefficient at the boundary between two nonmagnetic, conducting media in terms of their complex permittivities.

Exercise 8-3: Obtain expressions for the average power densities in media 1 and 2 for the fields described by Eqs. (8.22a) through (8.23b), assuming medium 1 is slightly lossy with η_{c_1} approximately real.

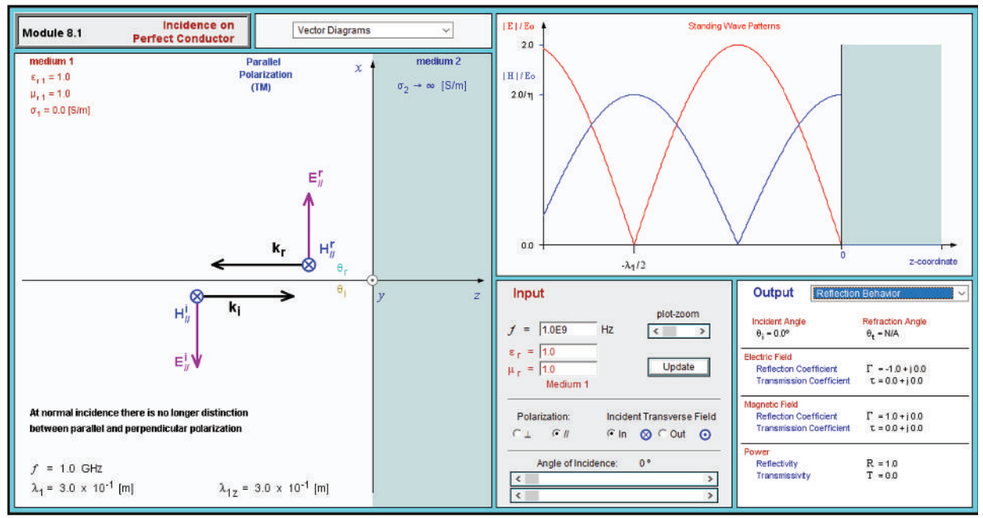
Answer: (See EM.)

$$\mathbf{S}_{av_1} = \hat{\mathbf{z}}\,\frac{|E_0^i|^2}{2\eta_{c_1}}\left(e^{-2\alpha_1 z} - |\Gamma|^2 e^{2\alpha_1 z}\right),$$

$$\mathbf{S}_{av_2} = \hat{\mathbf{z}}|\tau|^2\,\frac{|E_0^i|^2}{2}\,e^{-2\alpha_2 z}\,\mathfrak{Re}\left(\frac{1}{\eta_{c_2}^*}\right).$$

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模块 8.1 理想导体上的入射 观察由垂直入射到导体平面上的波与其反射相结合而产生的驻波图案。



概念问题 8-3：根据边界条件解释为什么 $\Gamma = -1$ 必须位于电介质和理想导体之间的边界处。

练习 8-1：为了消除正常入射平面波的反射，将厚度为 d 且相对介电常数为 ε_{r_2} 的电介质板插入两个相对介电常数为 $\varepsilon_{r_1} = 1$ 和 ε_{r_3} = 的半无限介质之间。 16. 使用四分之一波长变换器技术选择 d 和 ε_{r_2} 。假设 f = 3 GHz。

答案：$\varepsilon_{r_2} = 4$ 和 d = $(1.25 + 2.5n)$ (cm)，其中 n = 0, 1, 2, … 。 （参见EM。）

练习 8-2：用复介电常数表示两种非磁性导电介质边界处的法向入射反射系数。

答案：对于介质 1 (ε_1、μ_0、σ_1) 中的入射到介质 2 (ε_2、μ_0、σ_2) 中，

$$\Gamma = \frac{\sqrt{\varepsilon_{c_1}} - \sqrt{\varepsilon_{c_2}}}{\sqrt{\varepsilon_{c_1}} + \sqrt{\varepsilon_{c_2}}},$$

与 $\varepsilon_{c_1} = (\varepsilon_1 - j\sigma_1/\omega)$ 和 $\varepsilon_{c_2} = (\varepsilon_2 - j\sigma_2/\omega)$。 （参见EM。）

练习 8-3：获得方程 1 和 2 所描述的场中的平均功率密度表达式。 (8.22a) 到 (8.23b)，假设介质 1 略有损耗，且 η_{c_1} 近似为实数。

答案： （参见EM。）

$$\mathbf{S}_{av_1} = \hat{\mathbf{z}}\,\frac{|E_0^i|^2}{2\eta_{c_1}}\left(e^{-2\alpha_1 z} - |\Gamma|^2 e^{2\alpha_1 z}\right),$$

$$\mathbf{S}_{av_2} = \hat{\mathbf{z}}|\tau|^2\,\frac{|E_0^i|^2}{2}\,e^{-2\alpha_2 z}\,\mathfrak{Re}\left(\frac{1}{\eta_{c_2}^*}\right).$$

Technology Brief 15: Lasers

Lasers are used in CD and DVD players, bar-code readers, eye surgery, and multitudes of other systems and applications (**Fig. TF15-1**).

► A laser—acronym for ***L***ight ***A***mplification by ***S***timulated ***E***mission of ***R***adiation—is a source of ***monochromatic*** (single wavelength), ***coherent*** (uniform wavefront), narrow-beam light. ◄

This is in contrast with other sources of light (such as the sun or a light bulb), which usually encompass waves of many different wavelengths with random phase (incoherent). A laser source generating microwaves is called a ***maser***. The first maser was built in 1953 by Charles Townes, and the first laser was constructed in 1960 by Theodore Maiman.

Basic Principles

Despite its complex quantum-mechanical structure, an atom can be conveniently modeled as a nucleus (containing protons and neutrons) surrounded by a cloud of electrons. Associated with the atom or molecule of any given material is a specific set of ***quantized*** (discrete) ***energy states*** (orbits) that the electrons can occupy. Supply of energy (in the form of heat, exposure to intense light, or other means) by an external source can cause an electron to move from a lower energy state to a higher energy (***excited***) state. Exciting the atoms is called ***pumping*** because it leads to increasing the population of electrons in higher states (**Fig. TF15-2**(a)). ***Spontaneous emission*** of a photon (light energy) occurs when the electron in the excited state moves to a lower state (**Fig. TF15-2**(b)), and ***stimulated emission*** (**Fig. TF15-2**(c)) happens when an emitted photon "entices" an electron in an excited state of another atom to move to a lower state, thereby emitting a second photon of identical energy, wavelength, and wavefront (phase).

Principle of Operation

► Highly amplified stimulated emission is called ***lasing***. ◄

The lasing medium can be solid, liquid, or gas. Laser operation is illustrated in **Fig. TF15-3** for a ruby crystal surrounded by a flash tube (similar to a camera flash). A perfectly reflecting mirror is placed on one end of the crystal and a partially reflecting mirror on the other end. Light from the flash tube excites the atoms; some undergo spontaneous emission, generating photons that cause others to undergo stimulated emission; photons moving along the axis of the crystal bounce back and forth between the mirrors, causing additional stimulated emission (i.e., amplification), with only a fraction of the photons exiting through the partially reflecting mirror.

► Because all of the stimulated photons are identical, the light wave generated by the laser is of a single wavelength. ◄



Figure TF15-1 A few examples of laser applications.

技术简介 15：激光器

激光器用于 CD 和 DVD 播放器、条形码阅读器、眼科手术以及众多其他系统和应用（图 TF15-1）。

► 激光（受激辐射光放大的缩写）是单色（单波长）、相干（均匀波前）窄光束光源。 ◄

这与其他光源（例如太阳或灯泡）形成对比，其他光源通常包含具有随机相位（不相干）的许多不同波长的波。产生微波的激光源称为微波激射器。第一台微波激射器由 Charles Townes 于 1953 年制造，第一台激光器由 Theodore Maiman 于 1960 年制造。

基本原则

尽管原子具有复杂的量子力学结构，但可以方便地将原子建模为被电子云包围的原子核（包含质子和中子）。与任何给定材料的原子或分子相关的是电子可以占据的一组特定的量子化（离散）能态（轨道）。通过外部源提供能量（以热量、强光照射或其他方式）可以使电子从较低能态移动到较高能态（激发态）。激发原子称为泵浦，因为它会导致增加更高能态的电子数量（图 TF15-2(a)）。当处于激发态的电子移动到较低的状态时（图TF15-2（b））， 会发生光子（光能）的自发发射；当发射的光子“引诱”处于激发态的另一个原子的电子移动到较低的状态时，会发生受激发射（图TF15-2（c）），从而发射具有相同能量、波长和波前（相位）的第二个光子。

工作原理

► 高度放大的受激发射称为激光。 ◄

激光介质可以是固体、液体或气体。图 TF15-3 说明了被闪光灯管（类似于相机闪光灯）包围的红宝石晶体的激光操作。晶体的一端放置一面全反射镜，另一端放置一面部分反射镜。闪光灯管发出的光激发原子；有些发生自发发射，产生光子，导致其他发生受激发射；沿着晶体轴移动的光子在镜子之间来回反弹，引起额外的受激发射（即放大），只有一小部分光子通过部分反射镜射出。

► 由于所有受激光子都是相同的，因此激光器产生的光波具有单一波长。 ◄



图TF15-1 激光应用的几个示例。

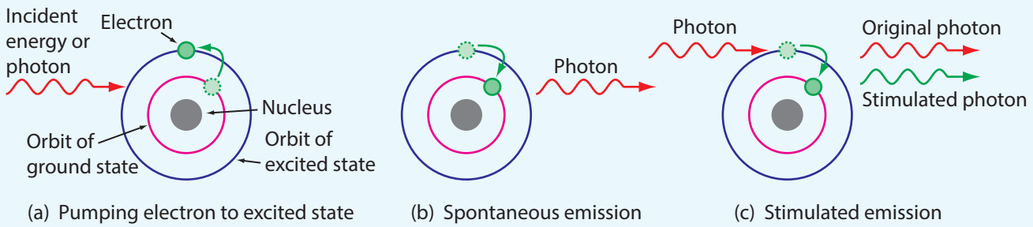


Figure TF15-2 Electron excitation and photon emission.

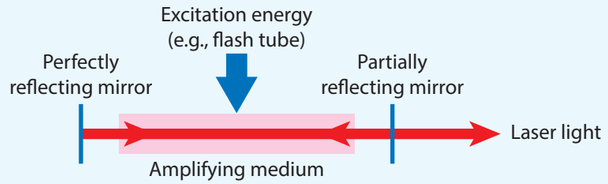


Figure TF15-3 Laser schematic.

Wavelength (Color) of Emitted Light

The atom of any given material has unique energy states. The difference in energy between the excited high energy state and the stable lower energy state determines the wavelength of the emitted photons (EM wave). Through proper choice of lasing material, monochromatic waves can be generated with wavelengths in the ultraviolet, visible, infrared or microwave bands.

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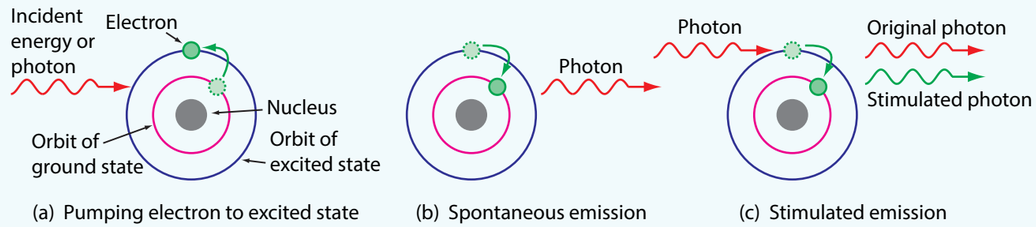


Figure TF15-2 Electron excitation and photon emission.

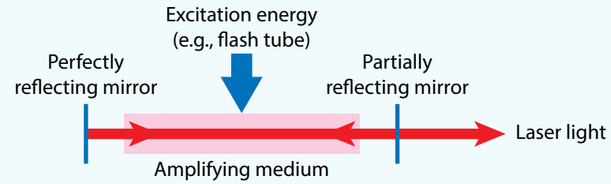


Figure TF15-3 Laser schematic.

发射光的波长（颜色）

任何给定材料的原子都具有独特的能态。激发的高能态和稳定的低能态之间的能量差

决定发射光子（电磁波）的波长。通过正确选择激光材料，可以产生波长在紫外线、可见光、红外线或微波波段的单色波。

8-2 Snell's Laws

In the preceding sections, we examined reflection and transmission of plane waves that are normally incident upon a planar interface between two different media. We now consider the oblique-incidence case depicted in **Fig. 8-9**, and for simplicity, we assume all media to be lossless. The $z = 0$ plane forms the boundary between media 1 and 2 with constitutive parameters (ε_1, μ_1) and (ε_2, μ_2), respectively. The two lines in **Fig. 8-9** with direction $\hat{\mathbf{k}}_i$ represent rays drawn normal to the wavefront of the incident wave, and those along directions $\hat{\mathbf{k}}_r$ and $\hat{\mathbf{k}}_t$ are similarly associated with the reflected and transmitted waves. The ***angles of incidence***, ***reflection***, and ***transmission*** (or ***refraction***), which are defined with respect to the normal to the boundary (the z axis), are θ_i, θ_r, and θ_t, respectively. These three angles are interrelated by ***Snell's laws***, which we derive shortly by considering the propagation of the wavefronts of the three waves. Rays of the incident wave intersect the boundary at O and O'. Here A_iO represents a constant-phase wavefront of the incident wave. Likewise, A_rO' and A_tO' are constant-phase wavefronts of the reflected and transmitted waves, respectively (**Fig. 8-9**). The incident and reflected waves propagate in medium 1 with the same phase velocity $u_{p_1} = 1/\sqrt{\mu_1\varepsilon_1}$, while the transmitted wave in medium 2 propagates with a velocity $u_{p_2} = 1/\sqrt{\mu_2\varepsilon_2}$. The time it takes for the incident wave to travel from A_i to O' is the same as the time it takes for the reflected wave to travel from O to A_r, and also the time it takes the transmitted wave to travel from O to A_t. Since time equals distance divided by velocity, it follows that

$$\frac{\overline{A_iO'}}{u_{p_1}} = \frac{\overline{OA_r}}{u_{p_1}} = \frac{\overline{OA_t}}{u_{p_2}}. \tag{8.26}$$

From the geometries of the three right triangles in **Fig. 8-9**, we deduce that

$$\overline{A_iO'} = \overline{OO'}\sin\theta_i, \tag{8.27a}$$

$$\overline{OA_r} = \overline{OO'}\sin\theta_r, \tag{8.27b}$$

$$\overline{OA_t} = \overline{OO'}\sin\theta_t. \tag{8.27c}$$

Use of these expressions in Eq. (8.26) leads to

$$\theta_i = \theta_r, \quad \textbf{(Snell's law of reflection)} \tag{8.28a}$$

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{u_{p_2}}{u_{p_1}} = \sqrt{\frac{\mu_1\varepsilon_1}{\mu_2\varepsilon_2}}. \tag{8.28b}$$

(Snell's law of refraction)

► ***Snell's law of reflection*** states that the angle of reflection equals the angle of incidence, and ***Snell's law of refraction*** provides a relation between $\sin\theta_t$ and $\sin\theta_i$ in terms of the ratio of the phase velocities. ◄

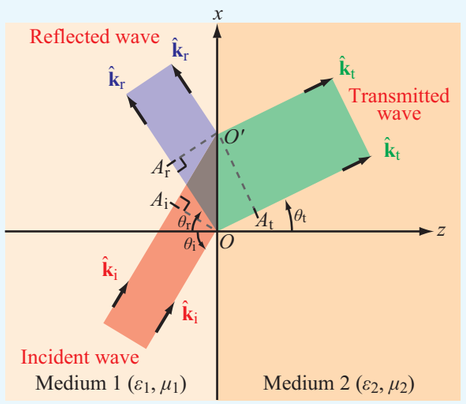


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

The ***index of refraction*** of a medium, n, is defined as the ratio of the phase velocity in free space (i.e., the speed of light c) to the phase velocity in the medium. Thus,

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r\varepsilon_r}. \tag{8.29}$$

(index of refraction)

In view of Eq. (8.29), Eq. (8.28b) may be rewritten as

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\mu_{r_1}\varepsilon_{r_1}}{\mu_{r_2}\varepsilon_{r_2}}}. \tag{8.30}$$

For nonmagnetic materials, $\mu_{r_1} = \mu_{r_2} = 1$, in which case

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}}} = \frac{\eta_2}{\eta_1} \quad (\text{for } \mu_1 = \mu_2). \tag{8.31}$$

Usually, materials with higher densities have higher permittivities. Air, with $\mu_r = \varepsilon_r = 1$, has an index of refraction $n_0 = 1$. Since for nonmagnetic materials $n = \sqrt{\varepsilon_r}$, *a material is often*

8-2 斯涅尔定律

在前面的部分中，我们研究了垂直入射到两种不同介质之间的平面界面上的平面波的反射和传输。我们现在考虑图8-9中描述的斜入射情况，为了简单起见，我们假设所有媒体都是无损的。 $z = 0$ 平面分别形成具有本构参数（ε_1，μ_1）和（ε_2，μ_2）的介质1和2之间的边界。图 8-9 中方向为 $\hat{k}_i$ 的两条线表示垂直于入射波波前绘制的射线，而沿方向 $\hat{k}_r$ 和 $\hat{k}_t$ 的射线同样与反射波和透射波相关。相对于边界法线（z 轴）定义的入射角、反射角和透射角（或折射角）分别为 θ_i、θ_r 和 θ_t。这三个角度通过斯涅尔定律相互关联，我们通过考虑三个波的波前传播很快就得出了该定律。入射波的光线在 O 和 O′ 处与边界相交。这里A_iO表示入射波的恒定相位波前。同样，A_rO'和A_tO'分别是反射波和透射波的恒定相位波前（图8-9）。入射波和反射波在介质 1 中以相同的相速度 $u_{p_1} = 1/\sqrt{\mu_1\varepsilon_1}$ 传播，而透射波在介质 2 中以速度 $u_{p_2} = 1/\sqrt{\mu_2\varepsilon_2}$ 传播。入射波从 A_i 传播到 O′ 所需的时间与反射波从 O 传播到 A_r 所需的时间相同，透射波从 O 传播的时间也相同

到A_t。由于时间等于距离除以速度，因此得出

$$\frac{A_iO'}{u_{p_1}} = \frac{OA_r}{u_{p_1}} = \frac{OA_t}{u_{p_2}}. \tag{8.26}$$

根据图8-9中三个直角三角形的几何形状，我们推导出

$$A_iO' = OO'\sin\theta_i, \tag{8.27a}$$

$$OA_r = OO'\sin\theta_r, \tag{8.27b}$$

$$OA_t = OO'\sin\theta_t. \tag{8.27c}$$

在等式中使用这些表达式。 (8.26) 导致

$$\theta_i = \theta_r, \quad \textbf{(Snell's law of reflection)} \tag{8.28a}$$

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{u_{p_2}}{u_{p_1}} = \sqrt{\frac{\mu_1\varepsilon_1}{\mu_2\varepsilon_2}}. \tag{8.28b}$$

(Snell's law of refraction)

► 斯涅尔反射定律规定反射角等于入射角，斯涅尔折射定律提供了 $\sin\theta_t$ 和 $\sin\theta_i$ 之间以相速度之比表示的关系。 ◄

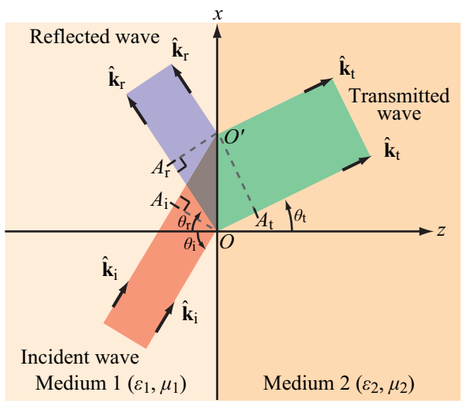


图 8-9 不同介质之间平面边界处的波反射和折射。

介质的折射率 n 定义为自由空间中的相速度（即光速 c）与介质中的相速度之比。因此，

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r\varepsilon_r}. \tag{8.29}$$

(index of refraction)

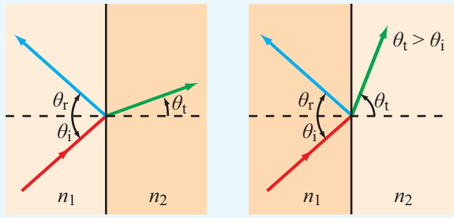
鉴于方程。 (8.29) ，等式。 (8.28b) 可以重写为

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\mu_{r_1}\varepsilon_{r_1}}{\mu_{r_2}\varepsilon_{r_2}}}. \tag{8.30}$$

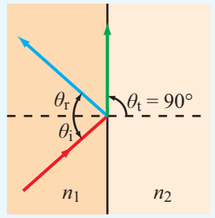
对于非磁性材料，$\mu_{r_1} = \mu_{r_2} = 1$，在这种情况下

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}}} = \frac{\eta_2}{\eta_1} \quad (\text{for } \mu_1 = \mu_2). \tag{8.31}$$

通常，密度较高的材料具有较高的介电常数。 $\mu_r = \varepsilon_r = 1$ 的空气的折射率 $n_0 = 1$。由于对于非磁性材料 $n = \sqrt{\varepsilon_r}$，材料通常是



(a) $n_1 < n_2$ (b) $n_1 > n_2$



(c) $n_1 > n_2$ and $\theta_i = \theta_c$

Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin\theta_t = (n_1/n_2)\sin\theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

referred to as more dense than another material if it has a greater index of refraction.

At normal incidence ($\theta_i = 0$), Eq. (8.31) gives $\theta_t = 0$, as expected. At oblique incidence $\theta_t < \theta_i$ when $n_2 > n_1$ and $\theta_t > \theta_i$ when $n_2 < n_1$.

► If a wave is incident on a more dense medium (**Fig. 8-10(a)**), the transmitted wave refracts inwardly (toward the z axis), and the opposite is true if a wave is incident on a less dense medium (**Fig. 8-10(b)**). ◄

A case of particular interest is when $\theta_t = \pi/2$, as shown in **Fig. 8-10(c)**; in this case, the refracted wave flows along the surface and no energy is transmitted into medium 2. The value of the angle of incidence θ_i corresponding to $\theta_t = \pi/2$ is called the ***critical angle*** θ_c and is obtained from Eq. (8.30) as

$$\sin\theta_c = \frac{n_2}{n_1}\sin\theta_t\Big|_{\theta_t=\pi/2} = \frac{n_2}{n_1} \tag{8.32a}$$

$$= \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.32b}$$

(critical angle)

If θ_i exceeds θ_c, the incident wave is totally reflected, and the refracted wave becomes a nonuniform ***surface wave*** that travels along the boundary between the two media. This wave behavior is called ***total internal reflection***.

Example 8-4: Light Beam Passing through a Slab

A dielectric slab with index of refraction n_2 is surrounded by a medium with index of refraction n_1, as shown in **Fig. 8-11**. If $\theta_i < \theta_c$, show that the emerging beam is parallel to the incident beam.

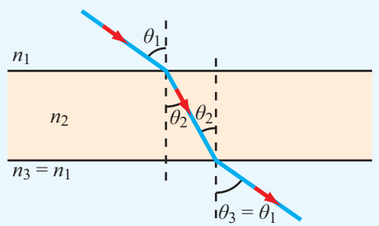


Figure 8-11 The exit angle θ_3 is equal to the incidence angle θ_1 if the dielectric slab has parallel boundaries and is surrounded by media with the same index of refraction on both sides (Example 8-4).

Solution: At the slab's upper surface, Snell's law gives

$$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1. \tag{8.33}$$

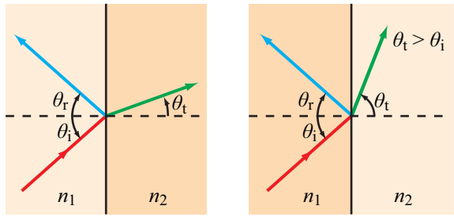
Similarly, at the slab's lower surface,

$$\sin\theta_3 = \frac{n_2}{n_3}\sin\theta_2 = \frac{n_2}{n_1}\sin\theta_2. \tag{8.34}$$

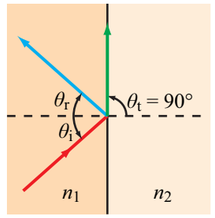
Substituting Eq. (8.33) into Eq. (8.34) gives

$$\sin\theta_3 = \left(\frac{n_2}{n_1}\right)\left(\frac{n_1}{n_2}\right)\sin\theta_1 = \sin\theta_1.$$

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(a) $n_1 < n_2$ (b) $n_1 > n_2$



(c) $n_1 > n_2$ and $\theta_i = \theta_c$

图 8-10 斯涅尔定律指出 $\theta_r = \theta_i$ 和 $\sin\theta_t = (n_1/n_2)\sin\theta_i$。如果 $n_1 < n_2$，则折射为 (a) 向内；如果 $n_1 > n_2$，则折射为 (b) 向外； (c)如果$n_1 > n_2$和θ_i等于或大于临界角$\theta_c = \sin^{-1}(n_2/n_1)$，则折射角为90°。

如果它具有更大的折射率，则被称为比另一种材料更致密。

在法线入射 ($\theta_i = 0$) 下，方程正如预期的那样，(8.31) 给出 $\theta_t = 0$。当 $n_2 > n_1$ 时倾斜入射 $\theta_t < \theta_i$，当 $n_2 < n_1$ 时倾斜入射 $\theta_t > \theta_i$。

► 如果波入射到密度较大的介质上（图8-10（a））， 则透射波向内折射（朝向z轴），如果波入射到密度较小的介质上则相反（图8-10（b））。◄

特别令人感兴趣的情况是当$\theta_t = \pi/2$时，如图8-10(c)所示；在这种情况下，折射波沿表面流动，没有能量传输到介质 2 中。对应于 $\theta_t = \pi/2$ 的入射角 θ_i 的值称为

临界角 θ_c 由方程式获得(8.30) 作为

$$\sin\theta_c = \frac{n_2}{n_1}\sin\theta_t\Big|_{\theta_t=\pi/2} = \frac{n_2}{n_1} \tag{8.32a}$$

$$= \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.32b}$$

(critical angle)

如果θ_i超过θ_c，则入射波被全反射，折射波成为沿着两种介质之间的边界传播的不均匀表面波。这种波行为称为全内反射。

例 8-4：光束穿过平板

折射率为n_2的电介质板被折射率为n_1的介质包围，如图8-11所示。如果$\theta_i < \theta_c$，则表明出射光束与入射光束平行。

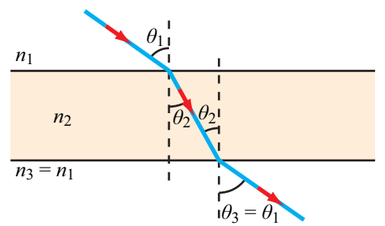


图 8-11 如果电介质板具有平行边界并且两侧都被折射率相同的介质包围，则出射角 θ_3 等于入射角 θ_1（示例 8-4）。

解：在板的上表面，斯涅尔定律给出

$$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1. \tag{8.33}$$

同样，在板的下表面，

$$\sin\theta_3 = \frac{n_2}{n_3}\sin\theta_2 = \frac{n_2}{n_1}\sin\theta_2. \tag{8.34}$$

代入方程式（8.33）代入等式。 (8.34) 给出

$$\sin\theta_3 = \left(\frac{n_2}{n_1}\right)\left(\frac{n_1}{n_2}\right)\sin\theta_1 = \sin\theta_1.$$